

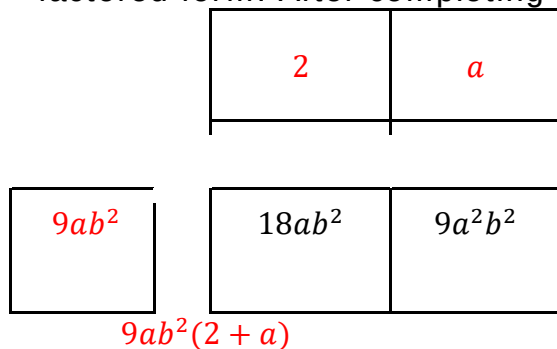
Grade 8
Unit 12
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

For questions 1-5, consider the monomial terms $18ab^2$ and $9a^2b^2$.

- Write all the factors that the coefficients share.
 $1, 3, 9$
- Write all the variables that the terms share.
 a, b^2
- Write all the common factors of the monomials. **Answers may vary.**
 $1, 3, 9, a, 3a, 9a, b, 3b, 9b, ab, 3ab, 9ab, b^2, 3b^2, 9b^2, ab^2, 3ab^2, 9ab^2$
- Complete the area model to find an equivalent expression to $18ab^2 + 9a^2b^2$ in factored form. After completing the area model, write the equivalent expression.

Answers may vary.

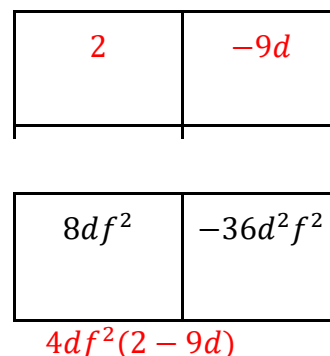
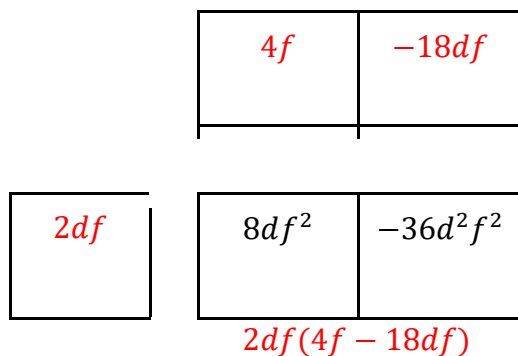


$9ab^2(2 + a)$

- Verify that the expression in question 4 is equivalent to $18ab^2 + 9a^2b^2$ using the distributive property.
 $9ab^2(2 + a)$
 $18ab^2 + 9a^2b^2$ ✓

For questions 6-7, complete the area models for $8df^2 - 36d^2f^2$ to find two equivalent expressions in factored form. After completing the area models, write the equivalent expressions. **Answers may vary.**

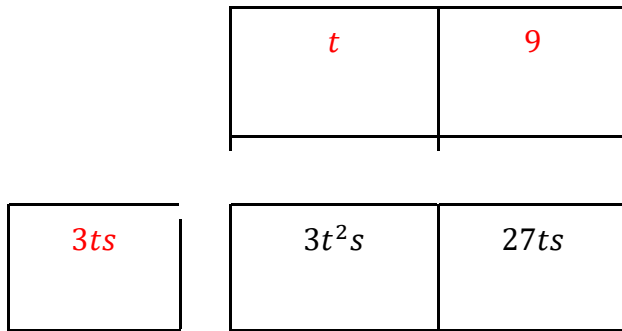
-
-
-
-
-
-
-



For questions 8-9, consider the expression $3t^2s + 27ts$.

8. Complete the area model to find an equivalent expression to $3t^2s + 27ts$ in factored form. After completing the area model, write the equivalent expression.

Answers may vary.



$$3ts(t + 9)$$

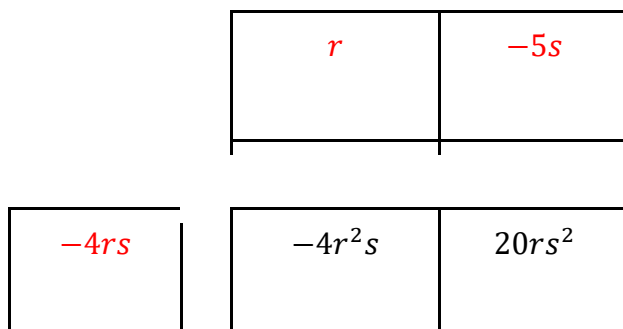
9. Verify that the expression in question 8 is equivalent to $3t^2s + 27ts$ using the distributive property.

$$3ts(t + 9)$$

$$3t^2s + 27ts$$

10. Complete the area model to find an equivalent expression to $-4r^2s + 20rs^2$ in factored form. After completing the area model, write the equivalent expression.

Answers may vary.



$$-4rs(r - 5s)$$